Computer Science & DA

Probability and Statistics

Joint P.M.F and P.D.F

Lecture No. 01

By- Dr. Puneet Sharma Sir

Recap of previous lecture

Topic Bivariate Random variable (Basics)

Topics to be Covered

Topic Joint p.m.f and p. d. f

JOINT P.M.F :

(*) if x_i & y_j are D.R.V then their joint P.M.F satisfies ;

(1) $P[X=x_i, Y=y_j] = P[x_i, y_j] \geq 0$

(2) $\sum_{i=1}^{m} \sum_{j=1}^{n} P(x_i, y_j) = 1$

(*) Individual Prob Dist. of X and Y are called Marginal Distribution.

(*) if $\boxed{P[x_i, y_j] = P(x_i) \cdot P(y_j)}$ $\forall$ i & j then X & Y are Independent R. Variables

(*) Conditional Distribution of X for given Y :

$$P\left[X=x / Y=y\right] = \frac{P[x,y]}{P(Y=y)}$$

& Conditional Dist of Y for given X is

$$P(Y=y / X=x) = \frac{P(x,y)}{P(X=x)}$$

[NAT]

#Q. A fair coin is tossed thrice. If X denote the number heads on the 1st toss and Y denotes the total number of heads then write it's joint p.m.f.

(ii) Also find marginal distributions of X and Y.

(iii) Evaluate $P(X = 1, Y \geq 1)$, $P(X \geq 0, Y \geq 2)$, $P(0 \leq X \leq 1, Y = 3)$

(iv) Conditional distribution of X when Y = 1

(v) Conditional distribution of Y when X = 1

(vi) $P(X = Y)$,

(vii) $P(1 \leq Y \leq 3) = ?$

(viii) $P(Y = 3) = ?$ $= \frac{1}{8}$

$S = \{(HHH)(HHT)(HTH)(HTT)(THH)(THT)(TTH)(TTT)\} \Rightarrow n(S) = 8$

$X = \{\text{No. of Heads in } 1^{st} \text{ toss}\} = \{0, 1\}$

$p_1 = P(X = 0H \text{ in } 1^{st} \text{ toss}) = \frac{4}{8}$

$p_2 = P(X = 1H \text{ in } 1^{st} \text{ toss}) = \frac{4}{8}$

X :	0	1
$P(X)$	$\frac{4}{8}$	$\frac{4}{8}$

$Y = \{\text{Total No. of Heads}\} = \{0, 1, 2, 3\}$

$p_1 = P(Y = 0H) = \frac{1}{8}$, $p_4 = P(X = 3H) = \frac{1}{8}$

$p_2 = P(X = 1H) = \frac{3}{8}$, $p_3 = P(X = 2H) = \frac{3}{8}$

Y :	0	1	2	3
$P(Y)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Now Joint Prob Dist is →

$X \backslash Y$	0	1	2	3	$P(X)$
0	1/8	2/8	1/8	0	4/8
1	0	1/8	2/8	1/8	4/8
$P(Y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

(ii) Marginal Distribution of X :-

X :	0	1
P(X)	$\frac{4}{8}$	$\frac{4}{8}$

& Marginal Dist of Y :-

Y :	0	1	2	3
P(Y) :	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(iii) $P(X=1, Y\geq 1) = ? = P(X=1 \cap Y = 1 \text{ or } 2 \text{ or } 3)$

$= P[(1,1) \text{ or } (12) \text{ or } (13)]$

$= \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{1}{2}$

$P[X\geq 0, Y\geq 2] = ? = P[(X=0 \text{ or } 1) \cap (Y=2 \text{ or } 3)]$

$= P[(02) \text{ or } (03) \text{ or } (12) \text{ or } (13)]$

$= \frac{1}{8} + 0 + \frac{2}{8} + \frac{1}{8} = \frac{4}{8}$

$P(0\leq X\leq 1, Y=3) = ? = P[(X=0 \text{ or } 1) \cap (Y=3)]$

$= P[(03) \text{ or } (13)] = 0 + \frac{1}{8} = \frac{1}{8}$

$P(X=Y) = ? = P[(00) \text{ or } (11)] = \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$

$P(1\leq Y\leq 3) = ?$ M-I $P(Y=1 \text{ or } 2 \text{ or } 3) = \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$

M-II $P[(X=0,1) \cap (Y=1,2,3)]$

$= P[(01) \text{ or } (02) \text{ or } (03) \text{ or } (11) \text{ or } (12) \text{ or } (13)] = 7/8$

(V) Conditional Dist of Y for given X=1 = ?

$$P(Y=0/X=1) = \frac{P(1,0)}{P(X=1)} = \frac{0}{4/8} = 0$$

$$P(Y=1/X=1) = \frac{P(1,1)}{P(X=1)} = \frac{1/8}{4/8} = \frac{1}{4}$$

$$P(Y=2/X=1) = \frac{P(1,2)}{P(X=1)} = \frac{2/8}{4/8} = \frac{2}{4}$$

$$P(Y=3/X=1) = \frac{P(1,3)}{P(X=1)} = \frac{1/8}{4/8} = \frac{1}{4}$$

So Req. Cond^n Dist is →

$Y=y$:	0	1	2	3
$P[Y=y/X=1]$:	0	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

(IV) Do yourself :-

$X=x$:	0	1
$P(X=x/Y=1)$:	$2/3$	$1/3$

#Q. Four coins are tossed, let x denotes numbers of heads and y denotes (number of Head-number of tails). Then write the joints pmf for such type of random experiment.

(i) Evaluate $P(-2 \le y < 4)$, (ii) $P(x \le 2, y \ge 2)$

(iii) $P(x \ge 3, -2 < y \le 4)$,

(iv) Conditional distribution of x given the value of y = 0

(v) Conditional distribution of y given the value of x = 3

(vi) Marginal distribution of x

(vii) Marginal distribution of y

(viii) $P(x = y) = ?$ (Ix) $p(x \le 2) = ?$ $= P(x=0,1,2) = \frac{1}{16} + \frac{4}{16} + \frac{6}{16} = \frac{11}{16}$

$S = \{(HHHH), (HHHT), \ldots, (HHTT), \ldots, (HTTT), \ldots, (TTTT)\} \Rightarrow n(S) = 2^4 = 16$ Quadruples

$X = \{\text{Number of Heads}\} = \{0, 1, 2, 3, 4\}$

X:	0	1	2	3	4
$P(X)$:	$\frac{1}{16}$	$\frac{{}^4C_1 = 4}{16}$	$\frac{{}^4C_2 = 6}{16}$	$\frac{{}^4C_3 = 4}{16}$	$\frac{1}{16}$

Now; $Y = \{\text{Number of Head} - \text{Number of Tail}\}$

$= \{-4, -2, 0, 2, 4\}$

Y:	-4	-2	0	2	4
$P(Y)$:	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Explanation. →

$P(Y=-4) = P[0H \,\&\, 4T] = \frac{1}{16}$

$P(Y=-2) = P(1H \,\&\, 3T) = \frac{4}{16}$

$P(Y=0) = P(2H \,\&\, 2T) = \frac{6}{16}$

$P(Y=2) = P(3H \,\&\, 1T) = \frac{4}{16}$

$P(Y=4) = P(4H \,\&\, 0T) = \frac{1}{16}$

Joint Distr.

$X \backslash Y$	-4	-2	0	2	4	$P(X)$
0	1/16	0	0	0	0	1/16
1	0	4/16	0	0	0	4/16
2	0	0	6/16	0	0	6/16
3	0	0	0	4/16	0	4/16
4	0	0	0	0	1/16	1/16
$P(Y)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	1

(ii) $P(-2 \le Y < 4) = P[Y = -2, 0, 2] = \frac{4}{16} + \frac{6}{16} + \frac{4}{16} = \frac{14}{16}$

$P(X \le 2, Y \ge 2) = ? = P[X = 0, 1, 2 \cap Y = 2, 4]$

$= P[(0\ 2) \text{ or } (0\ 4) \text{ or } (1\ 2) \text{ or } (1\ 4) \text{ or } (2\ 2) \text{ or } (2\ 4)]$

$= 0 + 0 + 0 + 0 + 0 + 0 = 0$

$P(X \ge 3, -2 < Y \le 4) = ? = P[X = 3, 4 \cap Y = 0, 2, 4]$

$= P[(3\ 0), (3\ 2), (3\ 4), (4\ 0), (4\ 2), (4\ 4)]$

$= 0 + \frac{4}{16} + 0 + 0 + 0 + \frac{1}{16} = \frac{5}{16}$

$P(X = Y) = ? = P[(0\ 0), (2\ 2), (4\ 4)] = 0 + 0 + \frac{1}{16} = \frac{1}{16}$

Conditional Dist of X for given $Y=0$ is

$$P\left[X=x / Y=0\right] = \frac{P[X=x \cap Y=0]}{P(Y=0)} \quad \text{--- (1)}$$

Here $P(Y=0) = \frac{6}{16}$

$$P(X=0/Y=0) = \frac{P(0,0)}{P(Y=0)} = \frac{0}{6/16} = 0$$

$$P(X=1/Y=0) = \frac{P(1,0)}{P(Y=0)} = \frac{0}{6/16} = 0$$

$$P(X=2/Y=0) = \frac{P(2,0)}{P(Y=0)} = \frac{6/16}{6/16} = 1$$

$$P(X=3/Y=0) = \frac{P(3,0)}{P(Y=0)} = \frac{0}{6/16} = 0$$

$$P(X=4/Y=0) = \frac{P(4,0)}{P(Y=0)} = \frac{0}{6/16} = 0$$

Hence Req. Conditional Dist is;

$X=x$:	0	1	2	3	4
$P(X=x/Y=0)$:	0	0	1	0	0

Similarly Conditional Dist of Y for given $X=3$ is

$Y=y$:	-4	-2	0	2	4
$P[Y=y/X=3]$:	0	0	0	1	0

$$P[Y=2/X=3] = \cancel{\frac{P[2,3]}{P(X=3)}} = ?$$

$$P(Y=2/X=3) = \frac{P(3,2)}{P(X=3)} = \frac{4/16}{4/16} = 1$$

#Q. Joint P.M.F. for random variables x and y are given in the table. Find $P[1 \le x < 3, y \ge 1] = ?$ $= P[x = 1,2 \cap y = 1,2] = P[(11),(12),(21),(22)]$

$\frac{1}{18} + \frac{1}{9} + \frac{1}{6} + \frac{1}{18} = \frac{7}{18}$

x \ y	0	1	2	P(x)
0	$\frac{1}{18}$	$\frac{1}{9}$	$\frac{1}{6}$	6/18
1	$\frac{1}{9}$	$\frac{1}{18}$	$\frac{1}{9}$	5/18
2	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{18}$	7/18
P(Y)	$\frac{6}{18}$	$\frac{6}{18}$	$\frac{6}{18}$	1

X: 0 1 2

P(X): $\frac{6}{18}$ $\frac{5}{18}$ $\frac{7}{18}$

Y: 0 1 2

P(Y): $\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$

#Q. Joint p.d.f of two discrete Random variables x and y is given as $f(x, y) = c\,(2x + y)$ where $0 \le x \le 2$ and $0 \le y \le 3$ then

(i) Find C

(ii) $P(x = 2, y = 1) = ?$

(iii) $P(x \ge 1, y \le 2) = ?$

x \ y	0	1	2	3	P(x)
0	0	c	2c	3c	6c
1	2c	3c	4c	5c	14c
2	4c	5c	6c	7c	22c
P(y)	6c	9c	12c	15c	42c

(i) w.k.that $42c = 1 \Rightarrow c = \frac{1}{42}$

(ii) $P(x=2, y=1) = ? = P(2,1) = 5c = \frac{5}{42}$

(iii) $P(x \ge 1, y \le 2) = ? = P[x = 1,2 \cap y = 0,1,2]$

$= P[(10), (11), (12), (20), (21), (22)]$

$= (2c + 3c + 4c) + (4c + 5c + 6c) = 24c$

$= \frac{24}{42}$

#Q. X and Y are two random variable having the joint density function,

$$\boxed{f(x,y) = \frac{1}{27}(2x+y)}$$

Where x and y can only be integer 0, 1 or 2. Find

(i) Marginal distributions of x and y

(ii) Conditional distribution of y given x = 1

X \ Y	0	1	2	P(X)
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
P(Y)	6/27	9/27	12/27	1

X :	0	1	2
P(X) :	$\frac{3}{27}$	$\frac{9}{27}$	$\frac{15}{27}$

Y :	0	1	2
P(Y) :	$\frac{6}{27}$	$\frac{9}{27}$	$\frac{12}{27}$

(ii) $P(Y=0/X=1) = \frac{P(1,0)}{P(X=1)} = \frac{2/27}{9/27} = \frac{2}{9}$

$P(Y=1/X=1) = \frac{3/27}{9/27} = \frac{3}{9}$

$P(Y=2/X=1) = \frac{4/27}{9/27} = \frac{4}{9}$

Y :	0	1	2
$P(Y=y/X=1)$:	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{4}{9}$

#Q. Two Dice are rolled simultaneously. Let x denotes the sum of two faces and y the absolute value of their difference. Find joint p.m.f for x and y.

HW

(i) Also evaluate P (x = y)

(ii) P [x = 7 ∩ y ≥ 3]

(iii) P [(x + y = 4) ∩ (x – y = 2)] = ?

(iv) Find marginal distributions of x and y

(v) Find conditional distribution of x given y = 2

$$S = \left\{ \begin{array}{cccccc} (11) & (12) & (13) & (14) & (15) & (16) \\ (21) & (22) & (23) & (24) & (25) & (26) \\ (31) & (32) & (33) & (34) & (35) & (36) \\ (41) & (42) & (43) & (44) & (45) & (46) \\ (51) & (52) & (53) & (54) & (55) & (56) \\ (61) & (62) & (63) & (64) & (65) & (66) \end{array} \right\} \Rightarrow n(S) = 36$$

$Y = \{\text{Modulus of their Difference}\}$
$= \{0, 1, 2, 3, 4, 5\}$

$X = \{\text{Sum of two faces}\} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

X :	2	3	4	5	6	7	8	9	10	11	12
P(X) :	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Y :	0	1	2	3	4	5
P(Y) :	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

$x \backslash y$	0	1	2	3	4	5	$p(x)$
2	1/36	0	0	0	0	0	1/36
3	0	2/36	0	0	0	0	2/36
4	1/36	0	2/36	0	0	0	3/36
5	0	2/36	0	2/36	0	0	4/36
6	1/36	0	2/36	0	2/36	0	5/36
7	0	2/36	0	2/36	0	2/36	6/36
8	1/36	0	2/36	0	2/36	0	5/36
9	0	2/36	0	2/36	0	0	4/36
10	1/36	0	2/36	0	0	0	3/36
11	0	2/36	0	0	0	0	2/36
12	1/36	0	0	0	0	0	1/36
$p(y)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	1

$x=5$
- (14) (41) $\Rightarrow y=3$
- (32) (23) $\Rightarrow y=1$

$x=6$
- (15)(51) $\Rightarrow y=4$
- (24)(42) $\Rightarrow y=2$
- (33) $\Rightarrow y=0$

$\frac{0}{1w}$

$x=7$
- (16)(61), (25)(52), (34)(43) $\Rightarrow$ $y=5$, $y=3$, $y=1$

$x=8$
- (26)(62), (35)(53), (44) $\Rightarrow$ $y=4$, $y=2$, $y=0$

THANK - YOU